

Electrical and Electronics Engineering and Robotics

ELEC345 High Speed Communications

Diversity Techniques

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Previously

- We looked at multipath propagation
- Talked about slow and fast fading and the effects of this on transmitted signals
 - Temporal fading
 - Frequency selective fading
 - Coherence time
 - Coherence bandwidth
- We looked at how Doppler can also cause fading.

This week

- We understand multipath but how do we mitigate against its effects?
 - We wish to send information from A to B in a mobile environment and be reasonably certain it will get there in the presence of multipath effects.
- Lets take a trip back in time.

Objectives for today:

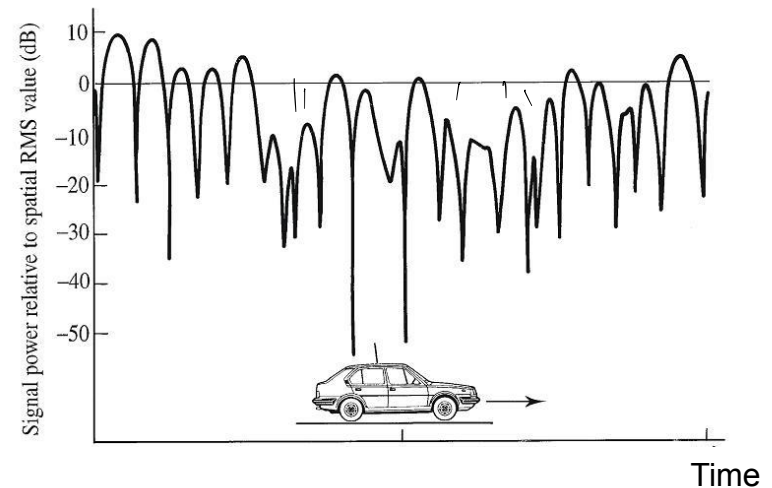
- To introduce the various forms of diversity reception and explain their advantages
- To describe the way in which diversity improvement is quantified
- To compare and explain the diversity spacing requirements at mobile terminals and base-stations

Mitigation of fast fading

- Is achieved by various forms of *diversity*
 - 1.
 - 2.
 - 3.
 - 4.
 - 5.
- All diversity techniques rely on the provision of two or more channels between which the fading correlation is low

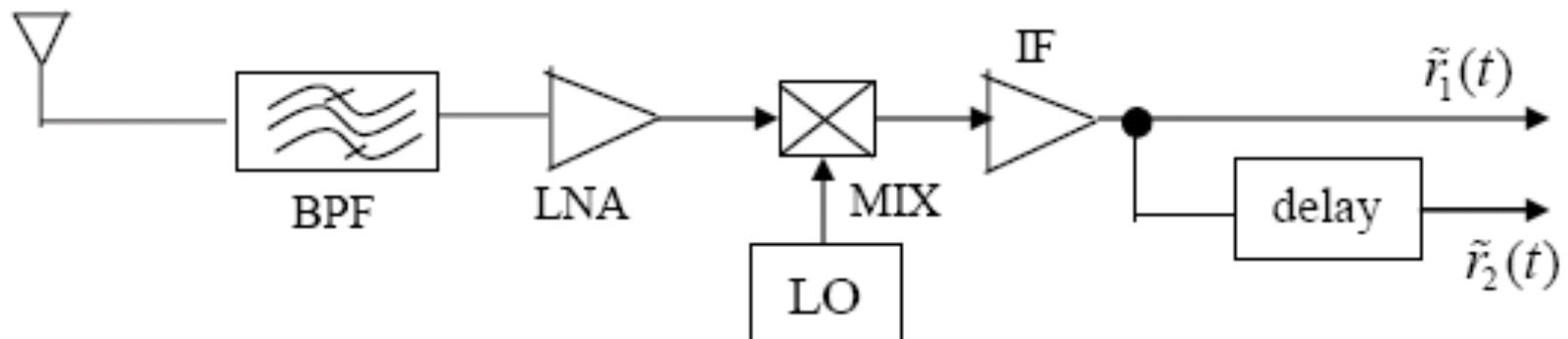
Time diversity

- Temporal fading of channels means fades sufficiently separated in time (at the same point in space) are de-correlated
- Retransmissions with this temporal spacing therefore provide diversity gain
- So related to the coherence time of the channel.
- Rarely used since:
 1. Reduces..
 2. It adds...



Time diversity

- With Doppler spread, the complex gain varies with time, so repeat the information after a few multiples of $(1/f_d)$. This is time diversity, (consumes extra transmission time).



Example

- A vehicle carrying a mobile terminal is receding from a 900 MHz GSM transmitter at 120 km/h. Calculate the Doppler shift in the line of sight component. In a time diversity technique, what would be the minimum time-diversity required for such a case, also comment on the disadvantage of this diversity technique?

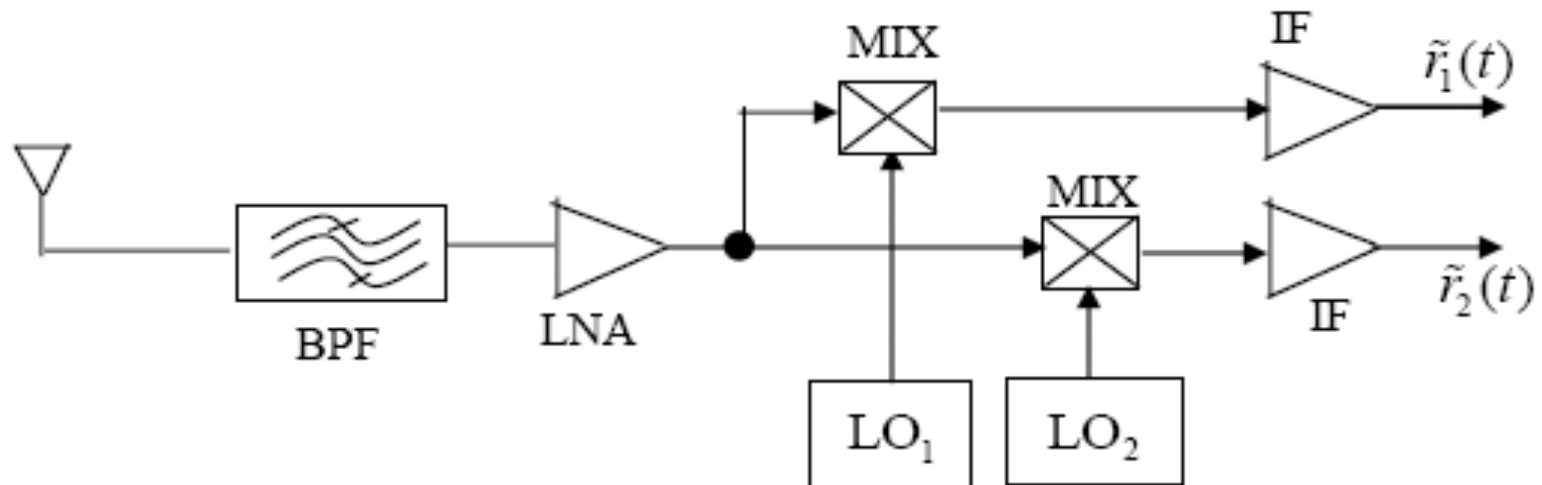
The Doppler shift is: $f_D = \frac{v}{c} f = \frac{120 \times 10^3 / 60^2}{3 \times 10^8} \times 900 \times 10^6 = 100 \text{ Hz}$

The time diversity required is a multiple of the inverse Doppler shift, so the minimum is

$$\text{TD} = \left(n \times \frac{1}{f_d} \right)_{\min. \rightarrow n=1} = 1 / 100 = 0.01 \text{ sec.} = 10 \text{ msec. or multiples of that.}$$

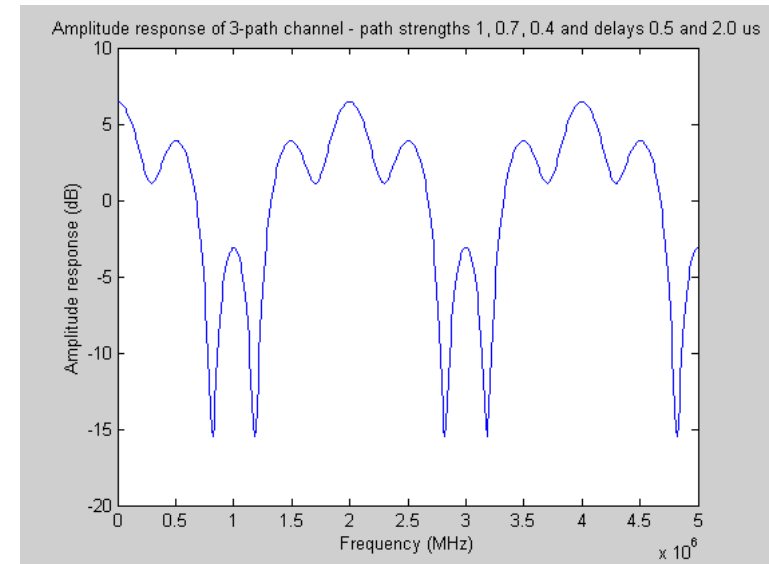
Frequency diversity

- With delay spread, frequencies farther apart than the coherence bandwidth experience independent values of frequency response. So we can transmit the same information on separate carriers. This is frequency diversity, (consumes extra bandwidth).



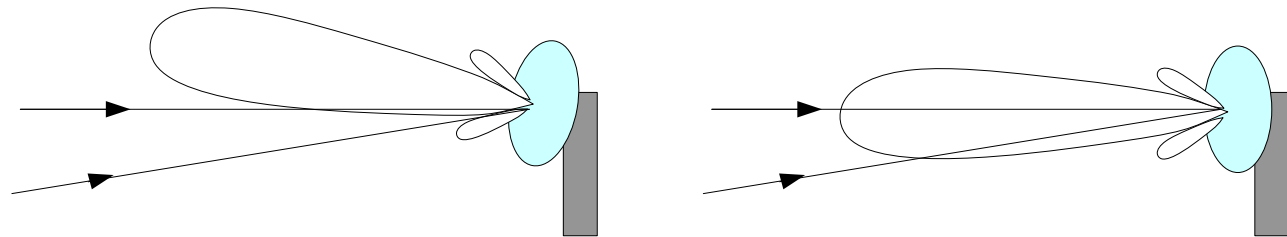
Frequency diversity

- Carriers (or channels) spaced by the decorrelation (coherence bandwidth) or more suffer uncorrelated fading and therefore give diversity gain
- System capacity reduced (by factor of 2)
- Not generally used in cellular applications



Angle Diversity

- Antennas with orthogonal (or quasi-orthogonal) radiation patterns respond to different subsets of multiple paths
- Fading therefore expected to be uncorrelated
- Not useful when omnidirectional antennas employed (as is the case at a mobile terminal)
- May be useful in principle for basestation diversity although diversity gain comes at expense of reduced power in each channel
- Used widely for microwave links
 - E.g. backhaul links in cellular systems



Angle diversity

Spatial Diversity

- For the classical model of a richly scattering environment
 - i.e many vertically polarised plane waves propagating in the horizontal plane incident on a vertically polarised omni-directional (non-directional) antenna
- The correlation between signal received by antennas spaced by ξ metres is: $R_{E_z}(\xi) = NE_0J_0(k\xi)$

where :

N is the number of paths

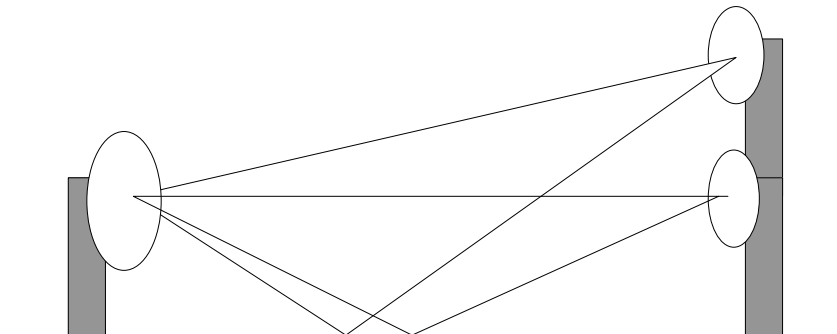
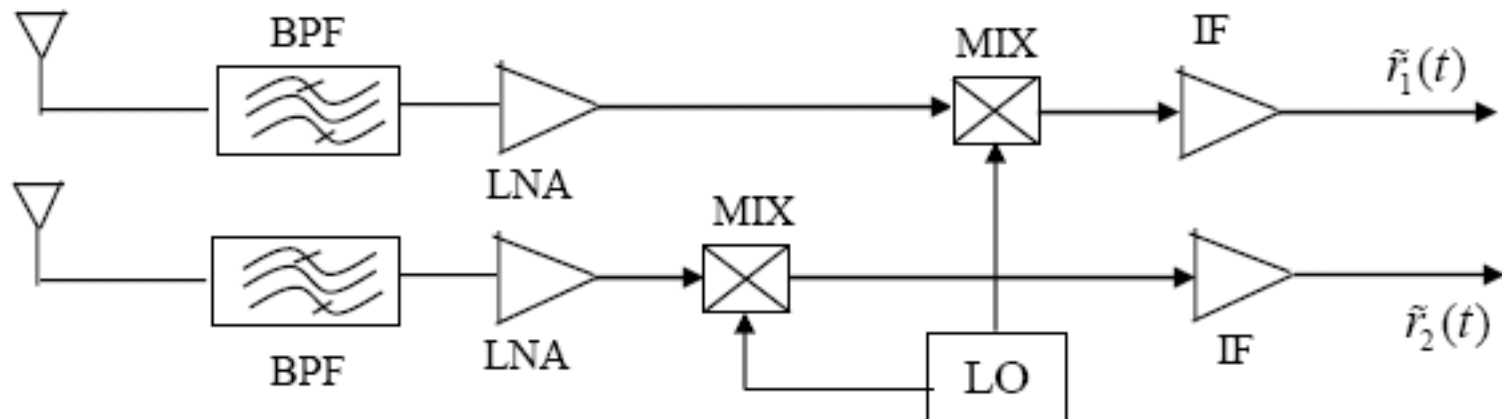
E_0 is the amplitude of the signal due to each path

J_0 is the zero order Bessel function of the first kind

k is the wave number (or spatial frequency) $= \frac{2\pi}{\lambda}$ [rad/m]

Space Diversity

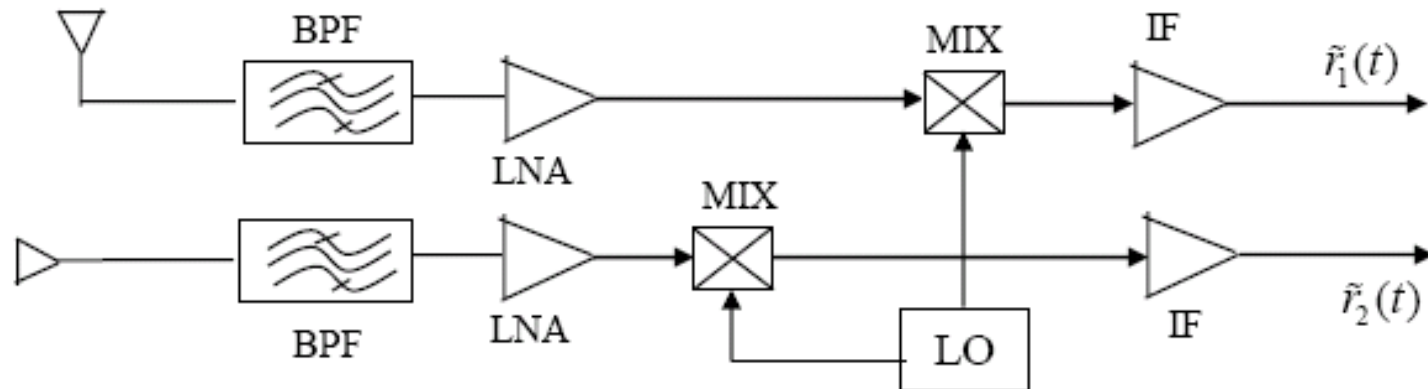
- With multiple receive antennas spaced far enough apart, we have space diversity. No need for extra transmission time or bandwidth (i.e. as in time and frequency diversity), each signal contributes extra energy. The costs are capital investment and physical space.



Height diversity

Polarization Diversity

- Correlation between orthogonally polarised fields at a single point in space is found to be low
- Allows co-location of diversity antennas
useful for small mobile handsets
- Sometimes combined with space diversity to reduce spacing needed for a given correlation coefficient (or diversity gain)



Diversity techniques to mitigate for multipath effects

1. Time
2. Frequency
3. Angle
4. Space
5. Polarization

Summary

- We've looked at a number of techniques to mitigate for multipath fading.
- We've looked at the advantages and disadvantages of each one.
- We've thought out how each one would be implemented

- Where next?
- ??
- ??

Any questions?